Data and Code Availability All code and data for demo and implementation is made available at: https://future.baulab.info

Acknowledgements This work was supported by Open Philanthropy and by the National Science Foundation (NSF) award 1901117. We thank the Center for AI Safety (CAIS) for making computing capacity available for this research.

References

Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. 2023. Eliciting latent predictions from transformers with the tuned lens.

Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramer, and Chiyuan Zhang. 2023. Quantifying memorization across neural language models. In *The Eleventh International Conference on Learning Representations*.

Nicholas Carlini, Chang Liu, Úlfar Erlingsson, Jernej Kos, and Dawn Song. 2019. The secret sharer: Evaluating and testing unintended memorization in neural networks. In *Proceedings of the 28th USENIX Conference on Security Symposium*, SEC'19, page 267–284, USA. USENIX Association.

Nicholas Carlini, Florian Tramer, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, Alina Oprea, and Colin Raffel. 2021. Extracting training data from large language models. In *USENIX Security Symposium*.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. BERT: pre-training of deep bidirectional transformers for language understanding. *CoRR*, abs/1810.04805.

Alexander Yom Din, Taelin Karidi, Leshem Choshen, and Mor Geva. 2023. Jump to conclusions: Short-cutting transformers with linear transformations.

Jeffrey L. Elman. 1990. Finding structure in time. *Cognitive Science*, 14(2):179–211.

Vitaly Feldman and Chiyuan Zhang. 2020. What neural networks memorize and why: Discovering the long tail via influence estimation. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS'20, Red Hook, NY, USA. Curran Associates Inc.

Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. 2020. The pile: An 800gb dataset of diverse text for language modeling.

Mor Geva, Avi Caciularu, Guy Dar, Paul Roit, Shoval Sadde, Micah Shlain, Bar Tamir, and Yoav Goldberg. 2022. LM-debugger: An interactive tool for inspection and intervention in transformer-based language models. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 12–21, Abu Dhabi, UAE. Association for Computational Linguistics.

Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. 2021. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 5484–5495, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Wes Gurnee, Neel Nanda, Matthew Pauly, Katherine Harvey, Dmitrii Troitskii, and Dimitris Bertsimas. 2023. Finding neurons in a haystack: Case studies with sparse probing.

Adi Haviv, Ido Cohen, Jacob Gidron, Roei Schuster, Yoav Goldberg, and Mor Geva. 2023. Understanding transformer memorization recall through idioms. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 248–264, Dubrovnik, Croatia. Association for Computational Linguistics.

Evan Hernandez, Arnab Sen Sharma, Tal Haklay, Kevin Meng, Martin Wattenberg, Jacob Andreas, Yonatan Belinkov, and David Bau. 2023. Linearity of relation decoding in transformer language models.

Daphne Ippolito, Florian Tramèr, Milad Nasr, Chiyuan Zhang, Matthew Jagielski, Katherine Lee, Christopher A. Choquette-Choo, and Nicholas Carlini. 2023. Preventing verbatim memorization in language models gives a false sense of privacy.

Michael I Jordan. 1997. Serial order: A parallel distributed processing approach. In *Advances in psychology*, volume 121, pages 471–495. Elsevier.

Shahar Katz and Yonatan Belinkov. 2023. Interpreting transformer's attention dynamic memory and visualizing the semantic information flow of gpt.

Jun Kong, Jin Wang, Liang-Chih Yu, and Xuejie Zhang. 2022. Accelerating inference for pretrained language models by unified multi-perspective early exiting. In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 4677–4686, Gyeongju, Republic of Korea. International Committee on Computational Linguistics.

Eric Lehman, Sarthak Jain, Karl Pichotta, Yoav Goldberg, and Byron C Wallace. 2021. Does bert pretrained on clinical notes reveal sensitive data? *arXiv preprint arXiv:2104.07762*.

Xiang Lisa Li and Percy Liang. 2021. Prefix-tuning: Optimizing continuous prompts for generation. *arXiv preprint arXiv:2101.00190*.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022a. Locating and editing factual associations in GPT. *Advances in Neural Information Processing Systems*, 36.

Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. 2022b. Mass editing memory in a transformer. *arXiv preprint arXiv:2210.07229*.

nostalgebraist. 2020. interpreting gpt: the logit lens.

Alec Radford, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners.

Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and Donald Metzler. 2022. Confident adaptive language modeling. In *Advances in Neural Information Processing Systems*.

Yixuan Su, Deng Cai, Yan Wang, David Vandyke, Simon Baker, Piji Li, and Nigel Collier. 2021. Non-autoregressive text generation with pre-trained language models. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 234–243, Online. Association for Computational Linguistics.

Jiuding Sun, Chantal Shaib, and Byron C Wallace. 2023. Evaluating the zero-shot robustness of instruction-tuned language models. *arXiv preprint arXiv:2306.11270*.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems*, 30.

Eric Wallace, Shi Feng, Nikhil Kandpal, Matt Gardner, and Sameer Singh. 2019. Universal adversarial triggers for attacking and analyzing nlp. *arXiv preprint arXiv:1908.07125*.

Ben Wang and Aran Komatsuzaki. 2021. Gpt-j-6b: A 6 billion parameter autoregressive language model.

Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed H. Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and William Fedus. 2022. Emergent abilities of large language models. *Transactions on Machine Learning Research*. Survey Certification.

Yisheng Xiao, Lijun Wu, Junliang Guo, Juntao Li, Min Zhang, Tao Qin, and Tie yan Liu. 2023. A survey on non-autoregressive generation for neural machine translation and beyond.

Ji Xin, Raphael Tang, Yaoliang Yu, and Jimmy Lin. 2021. BERxiT: Early exiting for BERT with better fine-tuning and extension to regression. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 91–104, Online. Association for Computational Linguistics.

Minjia Zhang and Yuxiong He. 2020. Accelerating training of transformer-based language models with progressive layer dropping.

A Appendix

Additional Figures

In this main paper, we report results based on models that are trained to optimize the $N = 1$ single token-ahead prediction, and we test those models for predictive accuracy for other N.

The same methods can also be used to optimize subsequent tokens, and the results of those methods are shown here. We find that optimizing for $N = 1$ works best and generalizes surprisingly well to other N, but that that optimizing for other N does not perform well for $N = 1$.

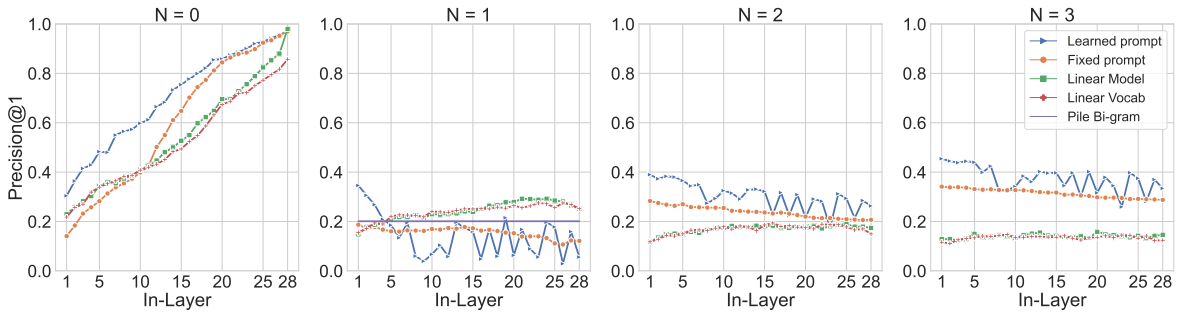

Figure 7: The Precision@1 (Accuracy) of all the methods trained with predicting the currently decoded token (teacher-forcing)

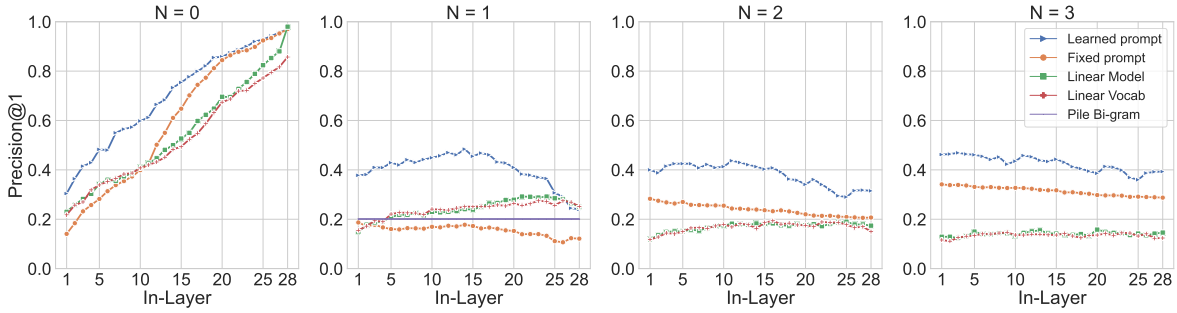

Figure 8: The Precision@1 (Accuracy) of all the methods trained with predicting the 1st next token

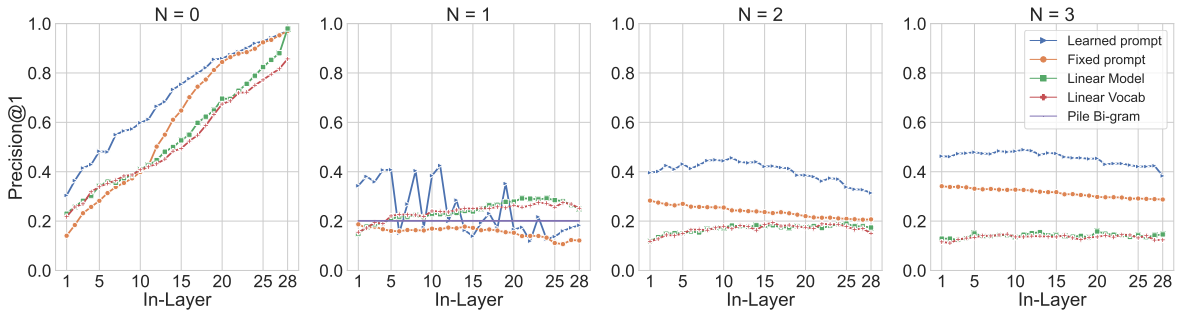

Figure 9: The Precision@1 (Accuracy) of all the methods trained with predicting the 2nd next token